RETHINKING ROAD RACING STANDARDS:

COMPARING NEW YORK ROAD RUNNERS RACE RESULTS

TO RUNNING INDUSTRY STANDARDS

by

JENNIE COUGHLIN

A master's capstone project submitted to the Graduate Faculty in Data Analysis and

Visualization in partial fulfillment of the requirements for the degree of Master of Science,

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APPROVAL

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This manuscript has been read and accepted for the Graduate Faculty in Data Analysis and Visualization in satisfaction of the capstone project requirement for the degree of Master of Science.

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ABSTRACT

Rethinking Road Racing Standards:

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Advisor: Timothy Shortell

Running has undergone a third boom in participation since 2020, driven largely by people seeking fitness options that were outside and socially distanced during the COVID-19 pandemic. Social media has also allowed runners from groups that did not traditionally participate to find community. As the road racing population has expanded to include more people, some road racing industry standards have not kept up. This project assesses what the road racing community looks like and measures industry standards against the community of participants to see what changes in those standards would be needed to make them inclusive for all participants.

To build the dataset, I scraped four years (2022-2025) of New York Road Runners race results at three distances: 5K, 4 miles and 10K. The years align with the general return to racing after the 2020 coronavirus pandemic as well as changes by NYRR in late 2022 to lengthen the course time limits for many races. The dataset includes more runners of color and runners with disabilities than most races, and the course time limits solve a typical problem with race results data, which is that slower runners are discouraged

from ever entering if they aren't fast enough to meet those limits. The analysis compares the population of NYRR finishers to several common running industry standards, including Couch to 5K, NYRR race corrals and RRCA guidelines for possible paces for running. It identifies places where the actual population of road racers diverges from those standards and recommends changes to include most or all of runners. A companion website presents the findings through a scrollytelling narrative, a series of charts and policy recommendations. The website is available at https://coughlinjennie.github.io/allrunners/ . All code, data and documentation are deposited at https://github.com/coughlinjennie/allrunners and at CUNY Academic Works.

ACKNOWLEDGEMENTS

I want to thank my thesis advisor, Timothy Shortell, for guiding this project from his feedback on the proposal in my first CUNY class through to the completion of this project. Professor Shortell showed me how to understand and build code to execute the analysis I had in mind and how to seek out data that would accurately represent a segment of a community that is often underrepresented in academic and other analysis. I also want to thank professors Abe Rubenstein and Ellie Frymire and my classmates in their data visualization classes for pushing me to develop my skills to translate analysis into visuals that are digestible to people encountering the project, as well as Professor Matt Gold and the rest of the faculty in the Data Analysis and Data Visualization program at the CUNY Graduate Center. The program was uniquely tailored to allow me to develop the skills I was seeking and the thinking to shape the project analysis, and it would not have been possible without every course I took.

I want to thank my colleagues at The New York Times, who worked with my unusual schedule while I was taking classes, especially my Audience team colleagues who occasionally had to cover for me when news broke while I was in class.

Finally, I want to thank Martinus Evans and Latoya Shauntay Snell. They allowed me to see and understand their mindset and approach to changing the running community to make it more inclusive when I was reporting a story for the Times in 2023 and provided the essential question that drove this project and many of my class projects during the pursuit of my degree: What does it look like when the running community allows everybody to belong?

DEDICATION

To Coaches Paul Trovato and Fran Bositis and in memory of Coaches Tom Geysen and Paul Davey for instilling the love of running in me, and to Ted Spiker and the Sub-30 Club for providing a space for all runners, no matter what pace or stage we're at.

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Zip file containing the contents of the GitHub repository at the time of deposit (https://github.com/coughlinjennie/roadracing)

A NOTE ON TECHNICAL SPECIFICATIONS

The data for the project is the NYRRclean.csv file, which can be loaded into your tool of choice. Analysis was done using R and RStudio.

The R notebook NYRRclean.Rmd details the work that was done to process the data from the original NYRR.csv file and requires the tidyverse package.

The R notebook NYRR.Rmd lays out the code used for the analysis. It requires these packages:

- ggplot2
- dplyr
- forcats
- tidyverse
- cowplot
- colorspace
- ggrepel
- tidyfit

The webpage was built using Observable Plot, D3, CSS and JavaScript

DATA DICTIONARY

Field/Function	Data Type	Description	Usage
Time	Numeric	Finishing time converted to seconds. To convert to minutes, divide by 60 and then take the residual as the seconds.	
Pace	Numeric	Calculated by dividing the Time field by the Distance field, presented in seconds. Can be converted to minutes by dividing by 60 and using the residual as the seconds.	Used to compare data across different distances and to industry standards that are based on pace
Distance		5K (5 kilometers), 4M (4 miles), 10K (10 kilometers), Half (13.1 miles)	Categorize races for comparison
Age	Numeric	Age as given in the race results	Used to categorize a record by Age Group
Age Group	Text	Standard 5-year age groups used in road racing for assessing performance of runners. Smaller races sometimes combine multiple age groups if there are too few runners. Here, runners 80 and older are grouped together and runners younger than 20 are grouped together.	Used to examine the impact of industry standards on runners of different ages.
Gender	Text	M for male, W for female, X for nonbinary	Used to analyze differences in the data by gender
RunnerID	Text	Created by combining the RaceId and finishing place (Overall field) for each runner.	Unique identifier for the dataset

THE APPROACH

Road racing is in theory among the most accessible sports available. It requires only sneakers in terms of specialized equipment and does not require any sport-specific location to train. However, it also has a history in the United States of excluding people who did not fit the sport's original participants: largely fast, white, able-bodied men.

Before 1960, U.S. women were banned for life from competition if they raced a distance longer than 200 meters.

The iconic photo from the 1967 Boston Marathon was Kathrine Switzer's then-boyfriend blocking BAA race director Jock Semple from forcibly removing Switzer from the race.

Ted Corbitt, often called "the father of American long distance running," was sometimes banned from track meets in high school and college when white athletes refused to run against him.

When Rick Hoyt wanted to run races, his father, Dick, pushed him in a wheelchair. But to run the Boston Marathon the first time, Dick had to hit the same qualifying standard as a runner who wasn't pushing his son in a wheelchair.

It wasn't until 2021 that New York Road Runners became the first World Marathon Major host to allow nonbinary runners to register for a race in their own category, and World Athletics has been rolling back access for transgender, nonbinary and intersex runners in recent years.

As a slow runner myself, I have seen that there are people at the back of the field in many road races who are participating even though the running industry standards don't include us in their definition of a runner. Before the pandemic, road racing participation had been falling since 2013, with demand for race registrations falling even as the supply increased.[1] However, during the pandemic, outdoor running while lockdowns were in place didn't negatively affect the health of runners.[2] As more and more people have taken up running, especially since the Covid-19 pandemic sent people looking for a solo, outdoors, socially-distanced fitness option, the road racing community has expanded.[3][4] However the running industry standards haven't changed at the same pace to reflect the full cohort of participants in the sport.

One of the first books I read for this program, "Data Feminism," has "consider context" as one of its seven principles. That principle asserts that "data are not neutral or objective. They are the products of unequal social relations, and this context is essential for conducting accurate, ethical analysis." [5]

Race results are the product of an industry that started when the sport was the province of fast, white men, and many of the parameters for races still reflect those origins. I have deliberately sought out a subset of race results that offset that tendency, and in the analysis, many of the policy questions on the table are ones that would counter the existing narratives within the industry. In that way, this project presents data with the context that is too often lacking when people just look at race results as a whole and draw conclusions

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[1] Kennedy, H., Baker, B. J., Jordan, J. S., & Funk, D. C. (2019). Running Recession: A Trend Analysis of Running Involvement and Runner Characteristics to Understand Declining Participation. Journal of Sport Management, 33(3), 215–228. https://doi-org.ezproxy.gc.cuny.edu/10.1123/jsm.2018-0261

[2] Kyra L.A. Cloosterman, Marienke van Middelkoop, Patrick Krastman, Robert-Jan de Vos. (2021). Running behavior and symptoms of respiratory tract infection during the COVID-19 pandemic: A large prospective Dutch cohort study. Journal of Science and Medicine in Sport, 24(4), 332-337. https://doi.org/10.1016/j.jsams.2020.10.009.

[3] Minsberg, Talya. The New York Times. Running From Coronavirus: A Back-to-Basics Exercise Boom. March 19, 2020. https://www.nytimes.com/2020/03/19/sports/running-exercise-coronavirus.html

[4] Minsberg, Talya. The New York Times. Running Boom Makes Race Bibs a 'Precious New York City Resource.' Oct. 1, 2025. https://www.nytimes.com/2025/10/01/nyregion/nyc-races-running-boom.html

[5] D'Ignazio, Catherine; Klein, Lauren F.. Data Feminism (Strong Ideas) (p. 149). MIT Press. Kindle Edition.

from them without considering who isn't included in the races because of choices made by organizers and providers of training resources.

By analyzing race results that have a long enough course limit to have a minimal upper bound for people who choose to participate in road races, we can move away from a definition of runner that is constrained by the race directors and organizations that provide training plans toward one where participants of all paces are providing the data that show what paces they are competing at in races of various distances. That shifts the discussion from who is able to count as a runner to one where runners can assess if a race organization or training plan includes them and advocate for change if it does not. Martinus Evans and Latoya Shauntay Snell are among the runners who have started to do that based on their lived experience, but this analysis could provide additional data to add weight to their efforts, in the same way that Christine Darden was able to work with data collected by Gloria Champine[6] to confirm her direct experience with discrimination at NASA.

Like Evans and Snell, I am a slow runner whose lived experience is often at odds with the industry definitions of who is a runner. When I did the Road Runners Club of America Level 1 coaching certification in 2019, during the morning of the first day, the trainer mentioned that research showed it was physically impossible for people to run slower than 13:45 minutes/mile. When I queried him about that during the lunch break, because my running pace was more in the 16:45-17:30/mile range for easy runs and 14:30 pace for races, his first response was "Oh, you mean run/walk."

I did not.

[6] D'Ignazio, Catherine; Klein, Lauren F.. Data Feminism (Strong Ideas) (p. 6). MIT Press. Kindle Edition.

So when I look at this data, my questions reflect my own experience as a slow runner and things I have heard from other runners who fall outside the traditional parameters of a runner, including runners I have interviewed for articles I have written for *The New York Times* on slow runners[7], runners with disabilities[8] and runners from groups that have not traditionally been included in the running community[9].

This project deliberately sought to source a dataset that includes many of the categories of runners who were not traditionally included in the sport's image. With that data, representative of the full spectrum of the road racing community, I then analyzed how many of those runners are excluded by various standards in the running industry:

- A common beginning training program.
- The way start corrals are structured to group runners with those of similar paces.
- The assumptions around running paces that guide course time limits and training programs.

This project includes recommendations for adjusting those standards to be inclusive of the existing road racing community. The data is also publicly available for people to use for their own analysis. I hope this project continues the discussion Evans, Snell and others have been leading about how the running industry can welcome all those who enjoy road racing.

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[7] Coughlin, Jennie. The New York Times. How Road Races Are Making Space for the Slowest Runners. May 13, 2023. https://www.nytimes.com/2023/05/13/sports/running-races-back-of-the-pack-slowest.html

[8] Coughlin, Jennie. The New York Times. Achilles Runners Confront Challenges Together. July 1, 2023 https://www.nytimes.com/2023/07/01/sports/achilles-international-running.html

[9] Coughlin, Jennie. The New York Times. Nontraditional Runners Are Finding Their Stride Online March 4, 2023. https://www.nytimes.com/2023/03/04/sports/running-clubs-online.html

LITERATURE SURVEY

The bulk of the literature out there around race times, results and large fields of runners tends to focus on faster runners, and on marathon or half-marathon races. Research that does testing of specific runners often defines the characteristics of runners with slower finishing times as a category, as in "Physiological and training characteristics of recreational marathon runners."[10] Even in that case it uses 4.5 hours or more as the slowest bucket for runners, well under the maximum for even the strict Boston Marathon course cutoff. One of the few studies available on 10K race participants and their finishing times stated that the authors used linear regression analysis "to examine the trends for age, sex, and finishing time for all participants completing the course in <1 hour."[11] Another one, which examined performance by half-marathon participants, required participants to be men, between 20 and 50, with half-marathon finishing times faster than 105 minutes.[12]

This project is deliberately looking at the people who have been typically excluded from studies because of either the researcher's choices, as in the Ogueta-Alday and Cushman papers, or because many races do have course limits that restrict participation and the analysis doesn't account for that. It also looks at three race distances: 5K, 4-mile and 10K. That will allow for the possibility that people participating in shorter races demonstrate differences from those in longer races such as the marathon and

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[10] Gordon, D., Wightman, S., Basevitch, I., Johnstone, J., Espejo-Sanchez, C., Beckford, C., … Merzbach, V. (2017). Physiological and training characteristics of recreational marathon runners. Open Access Journal of Sports Medicine, 8, 231–241. https://doi.org/10.2147/OAJSM.S141657

[11] Cushman, Dan M.1; Markert, Matthew2; Rho, Monica1. Performance Trends in Large 10-km Road Running Races in the United States. Journal of Strength and Conditioning Research 28(4):p 892-901, April 2014. | DOI: 10.1519/JSC.0000000000000249

[12] Ogueta-Alday A, Morante JC, Gómez-Molina J, García-López J (2018) Similarities and differences among half-marathon runners according to their performance level. PLoS ONE 13(1): e0191688. https://doi.org/10.1371/journal.pone.0191688

half-marathon. The data set is limited to races that have longer cutoff times. All of the races are hosted by New York Road Runners, based in New York City, which has specific programs encouraging older runners, youth runners and runners who can't afford race registrations to participate. NYRR also hosts the largest race that Achilles International partners on each year, and provides accommodations for runners with disabilities at all races, so it ensures that the Athletes With Disabilities (AWD) portion of the road racing community is represented as well.

Road races were largely suspended in 2020 and for much of 2021 because of the Covid-19 pandemic, so most of the literature was published before that time period, or uses data collected before the pandemic. This project also addresses that gap by focusing on races from 2022-2025.

DATA AND METHODS

A theoretical framework from my first CUNY class shaped this project and the data I used to address the questions I chose to explore. In Data Feminism,[13] Catherine D'Ignazio and Lauren F. Klein argue that data systems are embedded with the assumptions, priorities and power structures of the institutions behind them. To assess who was being excluded by running industry standards, I realized I needed to seek out a dataset that was not necessarily representative of the universe of road racers, but one that was more likely to include the runners who are traditionally excluded from the community and research around it. An added challenge is that road race results typically only collect age, gender and location, limiting the amount of demographic data available. Unless Athletes with Disabilities (AWD) are competing in a handcycle or wheelchair category, they are not designated as AWD in race results. Only a limited number of races provide gender options for registration beyond male and female, and the world governing body for the sport, World Athletics, is actively trying to exclude runners who don't fit specific criteria for gender. Finally, race time limits push slower runners to self-select out of the data by setting finish times that discourage them from registering for races.

The New York Road Runners dataset addressed a number of those issues. Because New York City's population is more racially and ethnically diverse than most of the country, the road racing community is also more diverse. The organization provides programs geared toward encouraging youth, senior citizens and low-income runners to participate. It allows runners to register for races in a non-binary category. It also partners with Achilles International, a nonprofit that serves athletes with disabilities, for the largest Achilles race

[13] D'Ignazio, Catherine; Klein, Lauren F.. Data Feminism (Strong Ideas) (p. 16). MIT Press. Kindle Edition.

in the United States each year, as well as providing AWD registration options for all races. NYRR is large enough and conducts enough races each year at varying distances to provide a robust data set in just the few post-2020 years. Finally, in late 2022, the organization lengthened the time limits for many of its races[14], and by the end of 2025, all the races included in this dataset had significantly longer time limits than pre-2020.

I built a series of Python scripts to scrape the race results data from the New York Road Runners results website, compiled them and calculated a pace field for each finish to allow for comparisons. I scraped the results of each 5K, 4-mile and 10K race in 2022 through 2025, a total of 459,812 records. I also used the age field to set an age group for each record to align with industry standards. Throughout the project, I will refer to finishes when referring to records rather than finishers because it is common for runners to run more than one NYRR race either in a year or over multiple years, so the number of people included in this dataset is lower than the number of records.

That raises a question: Does treating each runner as a unique person confound the analysis? In this case, I don't believe it makes a material difference to the questions under consideration. People can only run a race once, when defining a race as an event occurring on a specific date rather than as an event that occurs annually, so for any given field, a person can only appear once. For race organizers, there is typically no material difference between a runner who runs four races and four runners who each run one race. Both scenarios result in four race entries, and if those finish times are in the slower range of times we are examining, the impact on the race organization in terms of course limits and

[14] New York Road Runners. NYRR Time Cutoff Policy. July 31, 2026. https://www.nyrr.org/run/guidelines-and-procedures/policies-rules-and-regulations/nyrr-time-cutoff-policy

participation is the same. The largest difference would be if there were four individuals who were (or were not) able to sign up for a race because of how their expected finish times compared with the course time limit, the potential word of mouth as they discussed their ability (or inability) to register with friends and family and on social media could be much larger.

This analysis specifically does not attempt to differentiate between people running some or all of the race and those walking the race, so I am describing the people represented here as the road racing community. This is a deliberate choice, both to reflect the reality that people who complete a road race have chosen to sign up for it, regardless of the method of locomotion they use, and to respond to a common critique of slower finishers as "just walkers" or not "real" runners, the way the RRCA trainer assumed by my pace that I was alternating running and walking.

While each record has a pace field, I analyzed only the finishes at a given distance when comparing actual results to industry standards for specific distances, so only 5K finishes are used to compare the field to Couch to 5K training and only 10K finishes are used to compare the field to the corrals seeded by a 10K-equivalent pace.

NYRR also added a nonbinary category for all races in June 2021[15] and was one of the first major race organizations to do so. Many still do not. Sebastian Coe, president of World Athletics, said during a panel discussion at the 2023 New York City Marathon Expo that it is unlikely his organization, which governs international track and field competition, would add such a category.[16]

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[15] New York Road Runners. TCS New York City Marathon Announces Expanded Inclusivity Initiatives. Nov. 23, 2023. https://www.nyrr.org/media-center/press-release/20221017_tcsnycminclusivityinitiatives

[16] Said to New York Road Runners CEO Rob Simmelkjaer and written down by Coughlin, who was in the audience. Nov. 4, 2023.

At the time Coe said this, World Athletics, previously known as the International Association of Athletics Federations, prohibited runners with what they called “differences of sex development” from competing in women’s races at World Athletics-sanctioned events or being recognized for world records and world rankings unless they met stringent criteria.[17] The IAAF spent several years in court battles with runners, most visibly Caster Semenya of South Africa, over the issues of who was allowed to compete.

“For women like me, it is a complete ban under the guise of a new regulation,” Semenya said in her memoir.[18]

In the years since then, World Athletics has added further restrictions, requiring runners registering in the women’s division to undergo mandatory genetic testing[19] that the organization uses to determine their gender for the purposes of race eligibility. The International Olympic Committee, which Coe is a member of, has followed suit.[20]

“What is counted—like being a man or a woman—often becomes the basis for policymaking and resource allocation. By contrast, what is not counted—like being nonbinary—becomes invisible.” [21] There are only 1,661 nonbinary finishes in the NYRR dataset, so the conclusions that can be drawn from those are limited, but they at least exist,

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[17] World Athletics. World Athletics Eligibility Regulations for the Female Classification (Athletes with Differences in Sex Development). Version 3.0, approved by Council on March 23, 2023, and coming into effect on March 31, 2023.

[18] Semenya, Caster. The Race to Be Myself: A Memoir (p. 299). W. W. Norton & Company. Kindle Edition.

[19] World Athletics. World Athletics introduces SRY gene test for athletes wishing to compete in the female category. July 30, 2025. https://worldathletics.org/news/press-releases/sry-gene-test-athletes-female-category

[20] International Olympic Committee. Policy on the Protection of the Female (Women’s) Category in Olympic Sport and Guiding Considerations for International Federations and Sports Governing Bodies. March 26, 2026. https://stillmed.olympics.com/media/Documents/International-Olympic-Committee/EB/policy/policy-on-the-protection-of-the-female-category-english.pdf

[21] D'Ignazio, Catherine; Klein, Lauren F.. Data Feminism (Strong Ideas) (p. 97). MIT Press. Kindle Edition.

something that is uncommon in running industry data. Continuing to build out this dataset and doing more analysis with the nonbinary finishes data as more accumulates is one obvious additional research question for the future.

One limitation of this data set is there is no information on how experienced the runners are, and training gains compound over time.[22] Thus beginner runners who are the ones Couch to 5K is aimed at are likely to be slower than the field as a whole, but there is no way to measure that with this data set. Additional survey research in conjunction with a race organization to ask runners registering for races how much running experience they have would allow for improved data to better answer this question.

There is one additional caveat on this data set. Beyond the limitations on the data discussed above, there is also a potential portion of the running community who is not represented at all in this analysis: people who run but do not enter road races. That is another reason I have chosen to use road racing community rather than the running community when discussing the population measured here. There could be survey options for measuring and understanding the behavior of these non-racing runners, but they exceed the scope of this project because they, by definition, are not represented in race results. Instead, I will focus on people who are represented in race results.

[22] Jones A. M. (1998). A five year physiological case study of an Olympic runner. British journal of sports medicine, 32(1), 39–43. https://doi.org/10.1136/bjsm.32.1.39

INTRODUCTION

In 2017, Danish researchers released a report[23] showing American runners were the slowest they have ever been. They focused on understanding why runners were getting slower, but there is another question to answer: What does running look like now in terms of the range of paces that races and training resources should accommodate if their goal is to include the entire community? This analysis will provide answers that both industry groups and advocates for an inclusive running community can use to make races and training resources more inclusive of all road race participants.

The data that allows for this analysis is both simple and complicated: Race results. The results for many races are available online and readily accessible for viewing. However, there are a number of practices in the running community that affect that data and influence what it represents. The resources that are promoted in the running community can set expectations among runners for who "counts" as a runner.

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[23] Nikolova, Vania. American Runners Have Never Been Slower (Mega Study). Nov. 19, 2023. https://runrepeat.com/american-runners-have-never-been-slower-mega-study

The best-known beginner running program, Couch to 5K[24], is built around an assumption that running 30 minutes will allow you to complete a 5K, a 9:39 pace or faster. The program is even endorsed by the British National Health Service as an official exercise plan.[25] Across all the 5K races organized by NYRR between January 2022 and December 2025, the median pace for the 124,226 finishes was 9:30 and the average was just shy of 10:05.

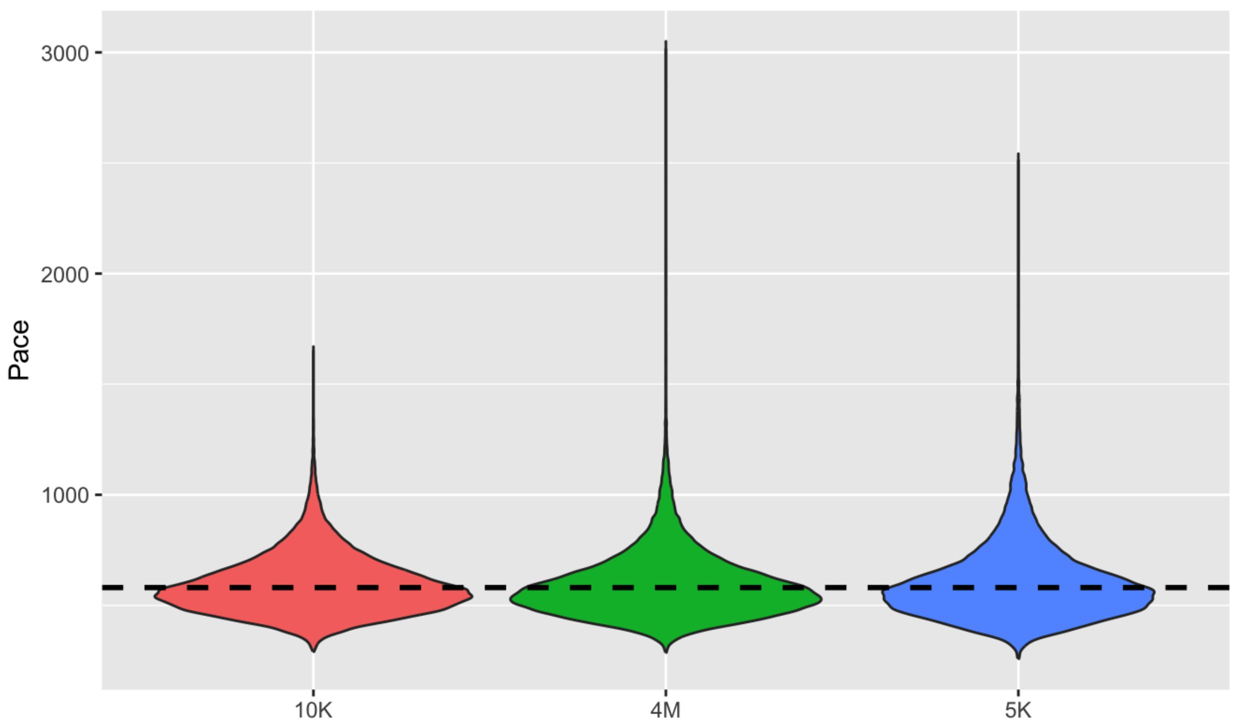

Figure 1: NYRR paces by race distance, with a line marking 9:39 pace, which produces a sub-30 5K race finish.

[24] Relph, N.; Taylor, S.L.; Christian, D.L.; Dey, P.; Owen, M.B. "Couch-to-5k or Couch to Ouch to Couch!?" Who Takes Part in Beginner Runner Programmes in the UK and Is Non-Completion Linked to Musculoskeletal Injury? Int. J. Environ. Res. Public Health 2023, 20, 6682. https://doi.org/10.3390/ ijerph20176682

[25] National Health Service. Get running with Couch to 5K. https://www.nhs.uk/better-health/get-active/get-running-with-couch-to-5k/

If a person does not register for a race because they do not believe they are able to run a 5K because they complete the existing Couch to 5K program and see that they have not run far enough, that alters the universe of people who sign up for a race. In those situations, the power to count the person as a runner whose performance can be measured by a race result was set by an industry expectation, not the runner's choice.

The 5K is the shortest distance race that NYRR hosts regularly. There are 22 5K races included in this data set, with 124,226 runners. When the NYRR data for 5K finishes is limited to the runners who registered in the women's division, the median pace rises to 10:12 and the average is just under 10:52 per mile.

Here are the 5K results by gender with a line marked at the 30-minute mark. You can see that the proportion of the men's field that finishes in under 30 minutes is markedly larger than the women's field or the non-binary field. The plots are not proportional to the size of the field, because the non-binary field would be too small to see if the plot was presented that way. The men's field is slightly larger (62,140 vs. 61,611) than the women's, but the gap is quite small given the total number of runners.

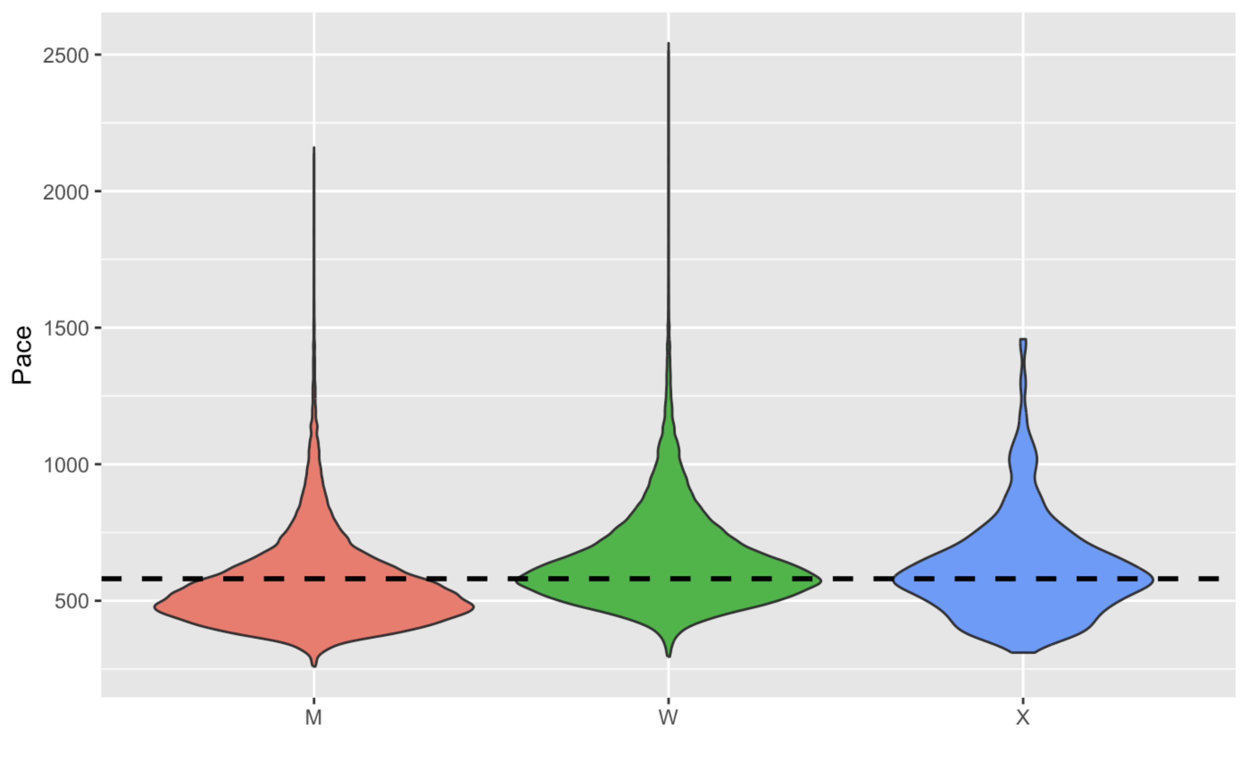

Figure 2: NYRR 5K race results by gender, with a line at the pace that produces a sub-30-minute finish.

This beginner program, created by a male runner,[26] doesn't meet the needs of the road racers who accounted for more than half of the women's finishes in NYRR 5K races during the four years measured.

There are simple fixes that could be made to the program that would make it more inclusive. If the training program was extended by a few weeks and took runners through 35 minutes, that covers up to the 75th percentile of all 5K finishes in NYRR races during the time period. Women are still under-represented in that cohort, with the finishing pace for the 75th percentile of women's finishes falling at 12:05, or a 37:32 finishing time.

[26] Clark, Josh. How Josh Clark Invented "Couch to 5K" and Helped Millions Start Running. Sept. 25, 2018. https://bigmedium.com/ideas/bbc-how-josh-clark-invented-couch-to-5k.html

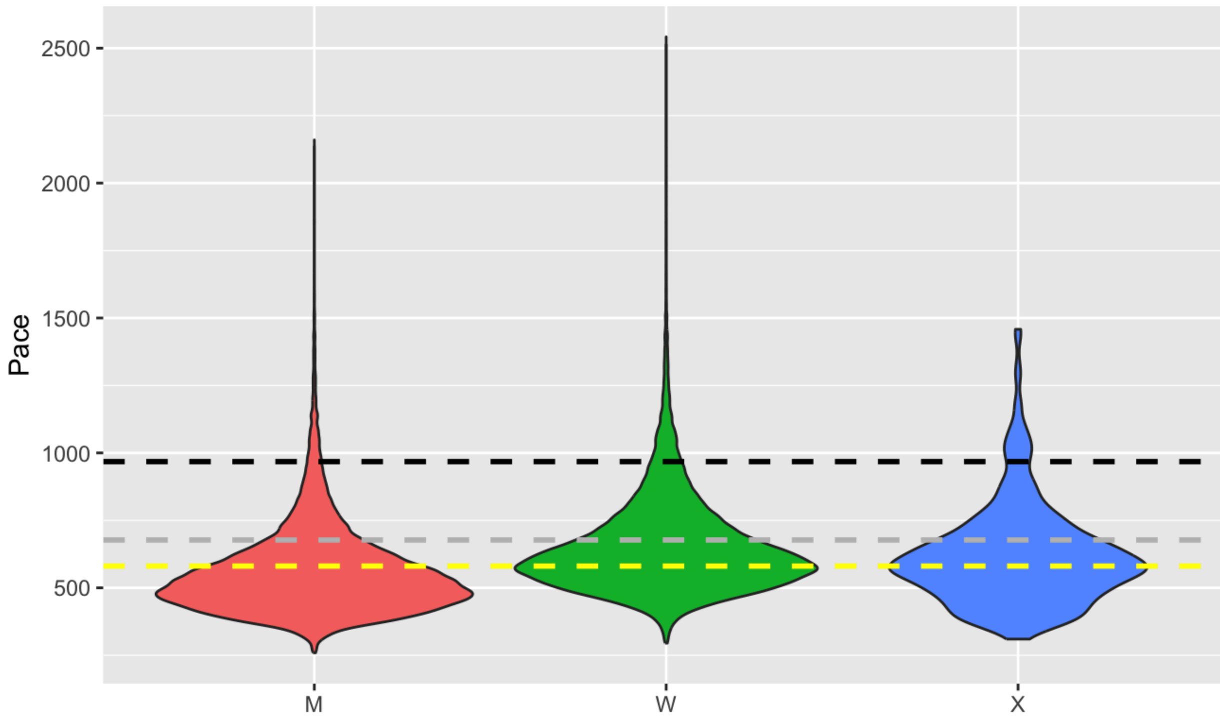

Figure 3: 5K finishes by gender, with lines at 30 minutes (yellow), 35 minutes (gray) and 50 minutes (black).

Using an outlier threshold of 1.5 times the interquartile range, the upper bound would be 16:10 pace for female finishes, or a 50:14 5K. Developing a beginner training program that took runners through 50 minutes, with specific spots where runners can assess if they are fast enough to race a 5K sooner than that, would meet the needs of a much larger segment of the road racing community.

The organization's senior running (Striders) and free community running (Open Run) programs are both oriented around the 5K distance. While we don't have a way to segment out beginning runners in this data set, we can look at runners by age. This is not to imply that all older runners are beginners. Those who began running as students or young adults during the first running boom that began in 1972 would be in their 70s in this dataset, so the older age groups likely also represent very experienced runners.

However, even experience doesn't offset the decline in speed that comes with age. When we break the 5K race results out by age group, the oldest age groups are notably slower than the age groups in the open category (younger than 40) and even among the younger master's (40 and older) runners.

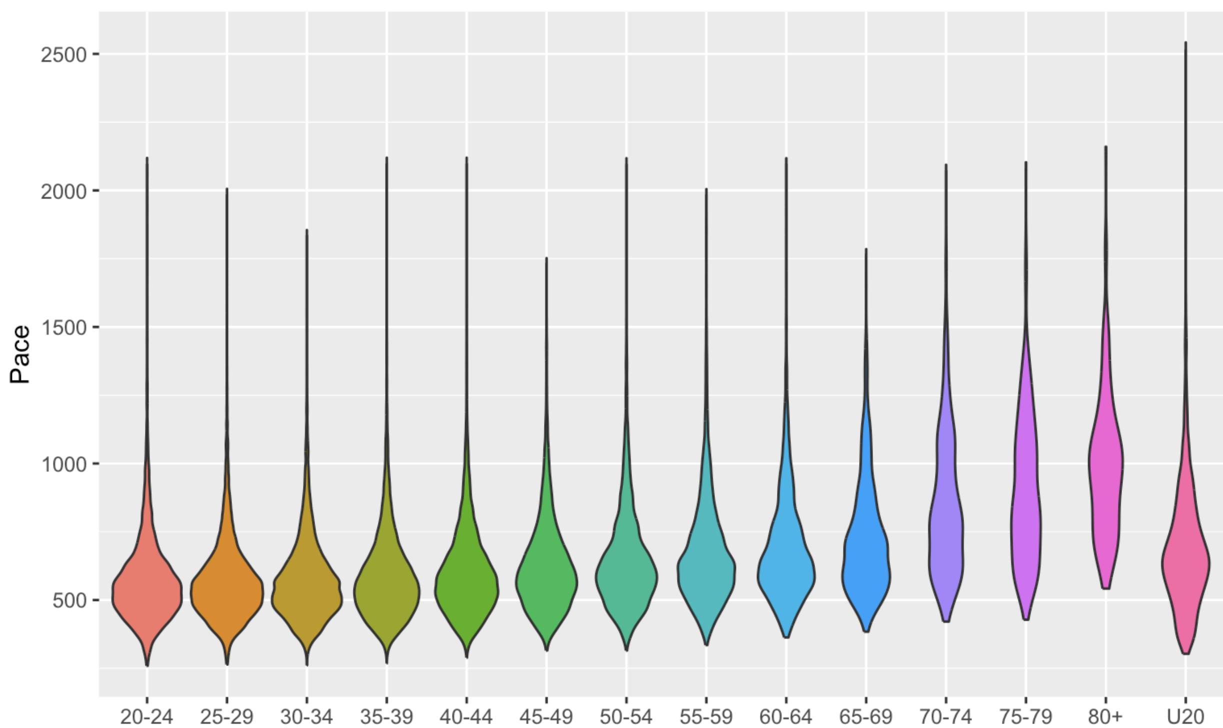

Figure 4: 5K finishes by age group, all genders

The age groups that include road racers 70 and older are the only ones with a substantial proportion of finishers that don't complete the race in less than 50 minutes. Training programs geared to seniors might need to consider going longer if they are geared to 5K completion. However, this is an area that would benefit from more research and collecting data that included information on experience levels. We can't tell from this data set if there are significant numbers of beginner runners older than 70 or if these are mostly longtime runners who have slowed with age. In the latter scenario, there is less need for a beginner-friendly program because these are experienced runners who are modifying their past training to accommodate age-related slowing.

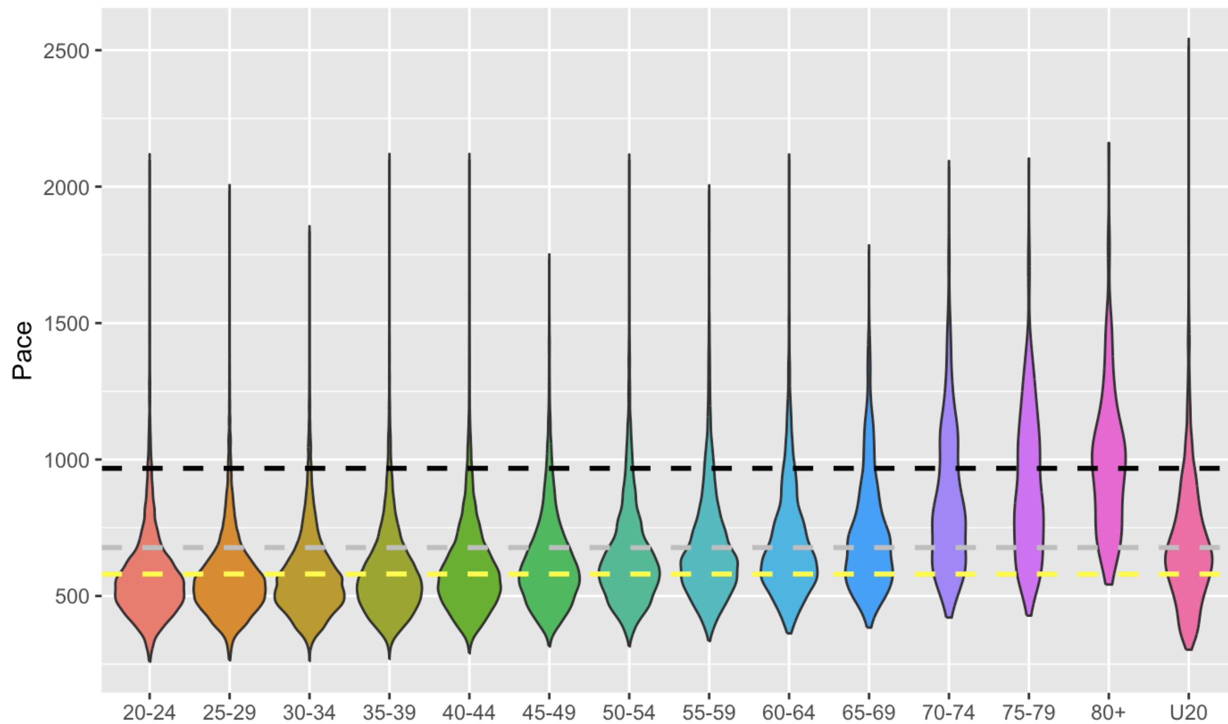

Figure 5: 5K finishes by age group, with lines at 30 minutes (yellow), 35 minutes (gray) and 50 minutes (black).

Whether these older runners are beginners who got involved through programs like Striders or longtime runners who are continuing to compete as they age, it's evident that they would benefit from longer course time limits and training plans that build to longer time on the course.

RACE ORGANIZATION

The larger the race, the more organization is required at the starting line. A small community race might just ask slower runners to start toward the back, while major marathons have waves and corrals seeded by expected finishing time or qualifying time. Because NYRR serves the largest city in the United States, even its smaller races have 3,000 to 5,000 runners, so it has a standard starting corral structure for weekly races[27]. A handful of races, including some Five Borough series races, are larger and use wave starts with a different corral structure. The organization gives all runners a “best pace” that it uses to organize them into corrals.[28] While people are allowed to move back to a slower corral, they are not allowed to start in a faster one. The “best pace” is set using the NYRR race result in the previous two years that equates to the fastest 10K race pace, and they publish a conversion table to allow runners to figure out what result would be required in a race that is a distance other than 10 kilometers to achieve a certain best pace.

The corrals are AA through L, with corrals AA and A having slightly different thresholds depending on the gender a runner registers as. Because of that, my analysis collapses those two corrals into a single group since the cutoff between A and B corrals is the first one that is the same no matter what gender the participant gave when registering. Corral L contains all runners with a best pace between 11:37 and 25:00, a range larger than the range of Corrals AA through K (4:00 to 11:36).

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[27] New York Road Runners. Pace Cuts. July 31, 2026. https://www.nyrr.org/run/guidelines-and-procedures/policies-rules-and-regulations/race-procedures#pace_cuts

[28] New York Road Runners. Best Pace and Corral Updates. July 31, 2026. https://www.nyrr.org/run/guidelines-and-procedures/policies-rules-and-regulations/best-pace-and-corral-updates

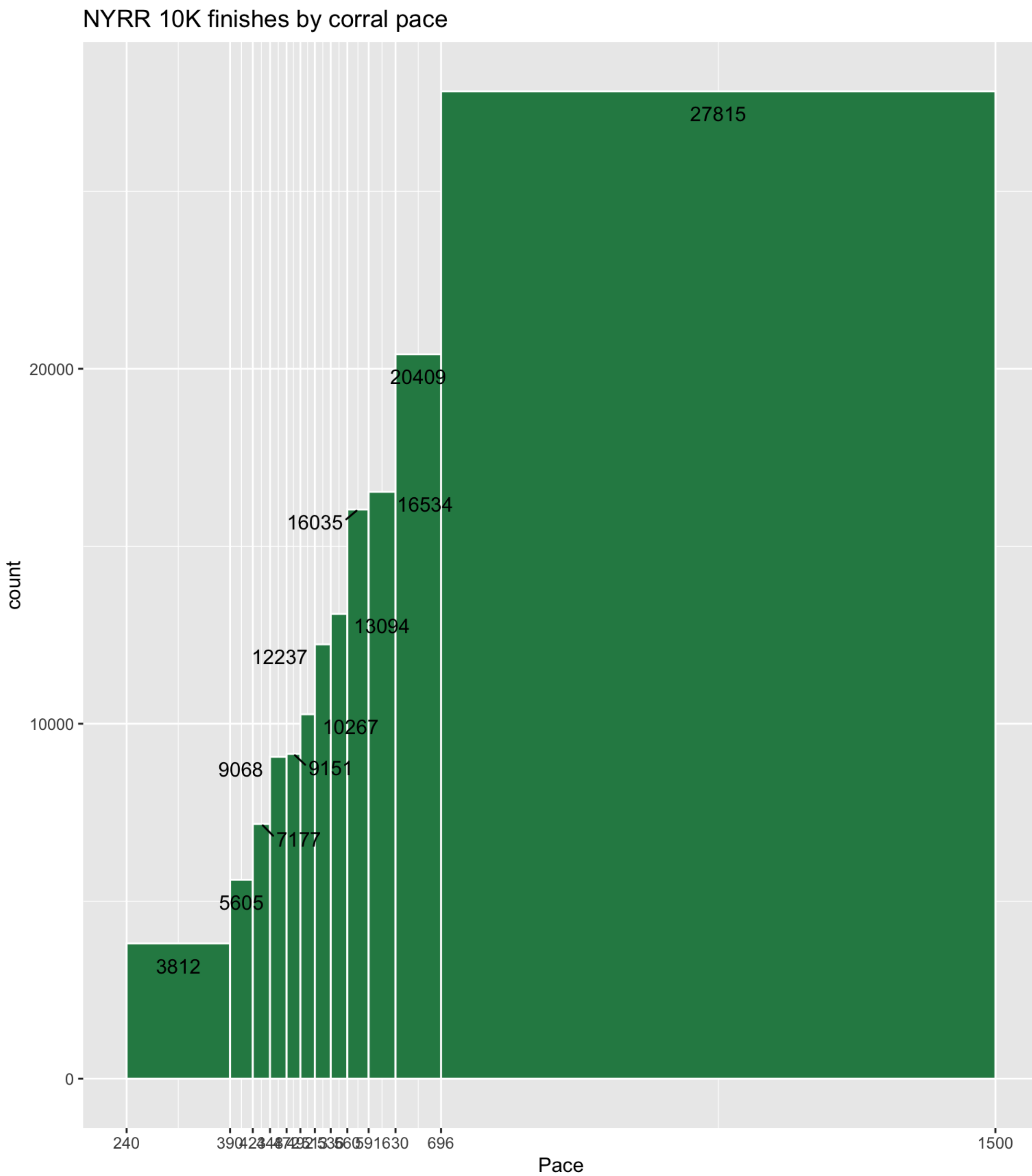

Figure 6: 10K finishes organized by start corral cutoffs used by NYRR

You can see that in each slower corral, more runners are placed there, with significantly more in the final corral than in the earlier ones. That corral actually has a slightly slower threshold than it did at one point. The cutoff between Corrals K and L used to be 11:30. Still, the notably higher count in that final bin suggests that the final corral might benefit from a different threshold, or the organization might benefit from adding one more corral. It also has by far the widest range of possible paces. In this data set, 19.5% (27,815) of the 10K finishes fall into Corral L. When we apply that to the typical weekly race, which has a field size of 3,000 to 8,000 participants, that puts between 585 and 1,560 runners in the final corral.

If the point of corrals is to ensure that runners are placed with people of roughly similar paces, placing almost 20 percent of runners into a single corral that has a range of paces that stretches more than 13 minutes/mile seems counterproductive to that, especially when the total field divided by the numbers of corrals suggests between 8 and 8.5 percent would be in each, which would work out to between 240 and 680 runners in a corral. Since these standards are fixed, that distribution is impossible to ensure for every race, or even for a large collection of races, but there is a lot of room between the current distribution and an ideal distribution.

Here's what the field looks like with equal-sized bins. Again, this combines Corrals AA and A:

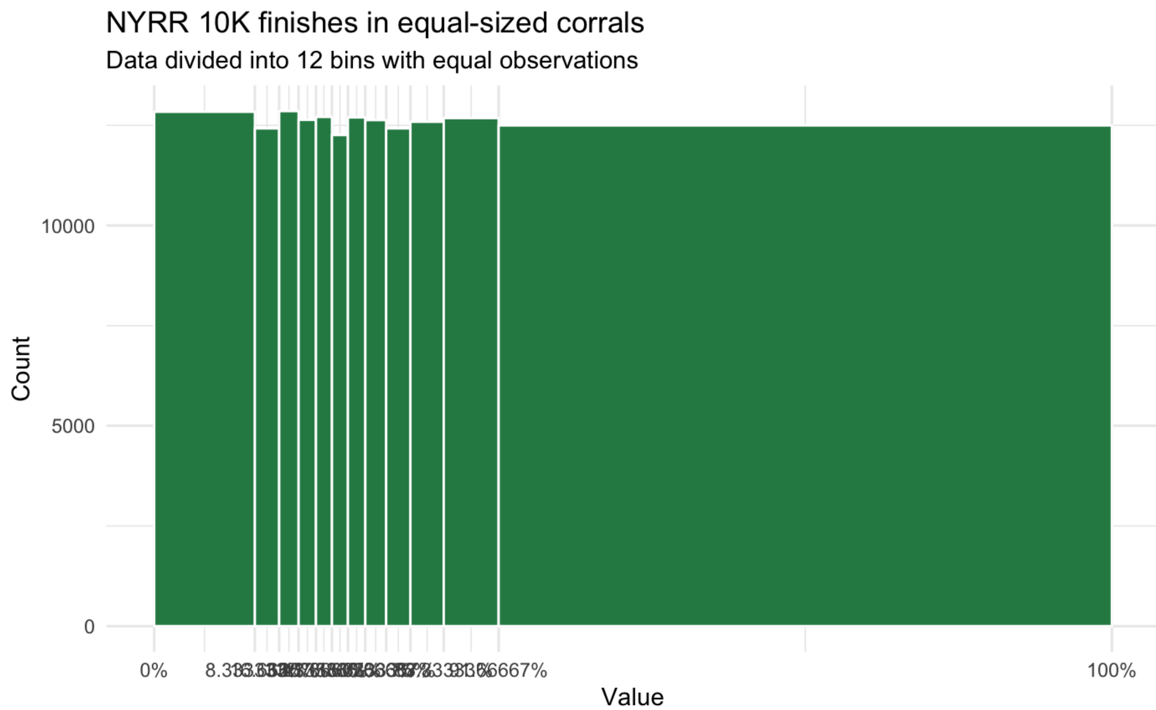

Figure 7: 10K finishes organized by start corral cutoffs using equal-sized bins

When the corrals are re-set to be equal distribution, with about 10,000 finishes in each corral, the tails are the most affected. The cutoff to get into the fastest corral gets much slower, dropping from 6:29 to 7:12, but after that the pace differential in each corral is pretty close, until the last few corrals.

Corral	Upper Bound	Range of paces
AA/A	7:15	3:15
B	7:51	36 seconds
C	8:19	28 seconds
D	8:44	25 seconds
E	9:07	23 seconds
F	9:30	23 seconds
G	9:55	25 seconds

H	10:25	30 seconds
I	11:00	35 seconds
J	11:48	48 seconds
K	13:07	1:19
L	25:00*	14:43

** The official cutoff is 25:00, but the full dataset calls for an upper bound of 27:50. Since it's the final corral, the actual pace doesn't matter much. Everybody who doesn't have a pace that puts them in Corral K is assigned to Corral L.*

Table 1: NYRR corrals parameters if they contained an equal number of runners, based on this dataset.

Using these pace breaks, far fewer people end up in the final corral. It doesn't solve the challenge of people of wildly different paces being grouped together, but it at least isn't cramming almost a fifth of the field into a single corral. One compromise solution would be to keep the first two corrals the same, then add Corrals M and N at the end, with the distribution for Corrals B-N being calculated using the above method.

Corral	Upper Bound	Range of paces
AA/A	6:29	2:29
B	7:12	42 seconds
C	7:47	35 seconds
D	8:13	26 seconds
E	8:36	23 seconds
F	8:58	22 seconds
G	9:20	22 seconds
H	9:41	21 seconds

I	10:06	25 seconds
J	10:35	29 seconds
K	11:10	35 seconds
L	11:58	48 seconds
M	13:15	1:17
N	25:00*	14:35

Table 2: NYRR corrals when the existing AA/A cutoffs are kept and two additional corrals are added, with an equal number of finishes in corrals B-N

The final corral continues to contain a large range of paces, but if you assume this equates to 7.69 percent of runners in each corral in a typical race, that means in a weekly race, which typically has 3,000 to 8,000 runners, that would put 230 to 615 runners in each corral, including the final one. Contrast that with the existing corral breaks, where a weekly race could have between 585 and 1,560 runners in the final corral.

Race organizers set the parameters for their races, including a course cutoff. This reflects how long they intend to leave the finish line open and record official finishers. On races with closed courses, roads are reopened at times that reflect this course time limit, and aid stations are similarly closed. Some racers require people to leave the course at that point, such as the Marine Corps Marathon's "Beat the Bridge" checkpoint[29]. Others, including the New York City Marathon, will state in information for the race[30] that runners must move to the sidewalk, but can continue along the course.

These times limit who is able to participate. When we examine [Berlin Marathon race results from 2000 to 2018](), two findings reflect the impact of course time limits.

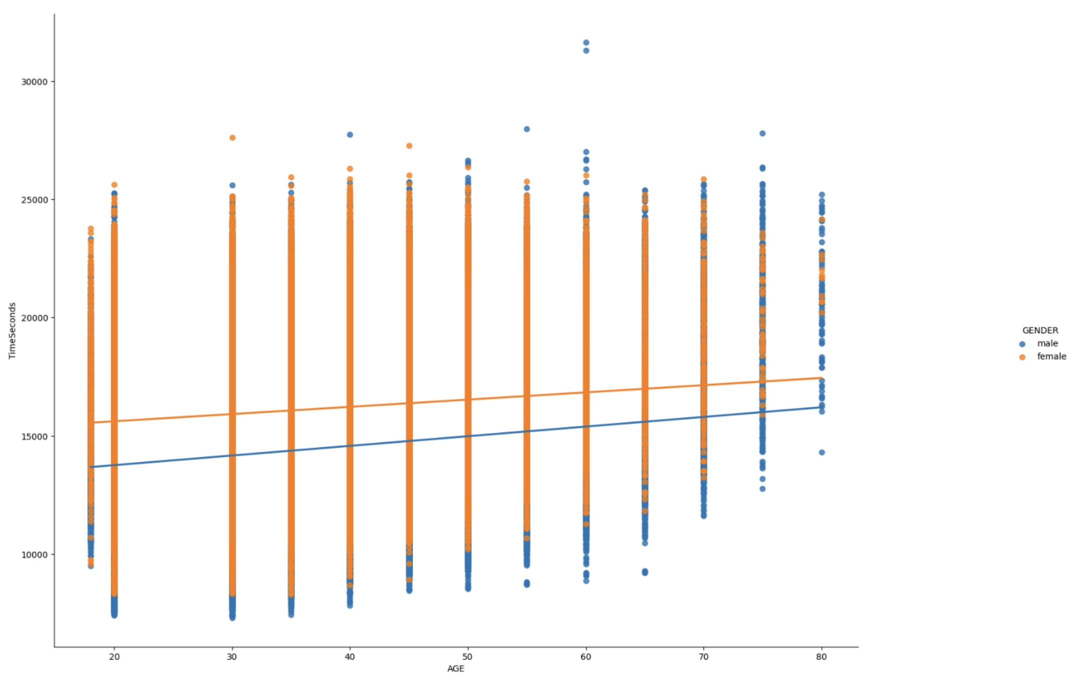

Figure 8: Berlin Marathon race results by age, 2000-2018

[29] Marine Corps Marathon Event Map > Gauntlets. Nov. 23, 2023. https://www.marinemarathon.com/event/marine-corps-marathon/map/

[30] New York Road Runners. TCS NYC Marathon Race Day > The Course > Sweep Buses and Bridge/Street Closures and Reopenings. Nov. 23, 2023. https://www.nyrr.org/tcsnycmarathon/race-day/the-course

When the Berlin results are analyzed by age group and gender, the fastest pace in each group declines with age, but the slowest runners in each age group all end in the same time range, which corresponds with the course time limit. One other finding, which we will return to, is that while the fastest runners in each age group are mainly men, the slowest runners are a mix of men and women.

Course time limits constrain the race results. Instead of reflecting the universe of people who choose to participate in road racing events, it reflects only those participants who choose to register for a particular race knowing the race parameters, including the time limit. Decisions made based on those race results would then exclude the people who self-selected out of that race, without ever noting that they had been excluded. These are people who know they can complete the distance, but not in the allotted time.

This data set was collected to avoid some of these pitfalls. Course time limits can vary widely, but the idea that 13:45 is a standard for what constitutes running does have some value, as 3:00 — the amount of time it takes to run 13.1 miles at 13:45 pace — is a common half-marathon time limit and many training plans for various distances top out with paces in that range. Likewise, that translates to a 6:00 marathon, and many marathons have 6:00 or 6:30 as a cutoff.

The NYRR data set doesn't include half-marathon data, but since pace tends to slow with longer distances, looking at the paces road racers are competing at for shorter distances should, if anything, underestimate how many people are excluded by the 13:45 standard. In the data collected, 33,734 finishes were at paces slower than 13:45, or 7.34 percent of the field. Further research that examines how this plays out at the half-marathon distance would allow more definitive assessment of the training resources available for the half-marathon and marathon race distances.

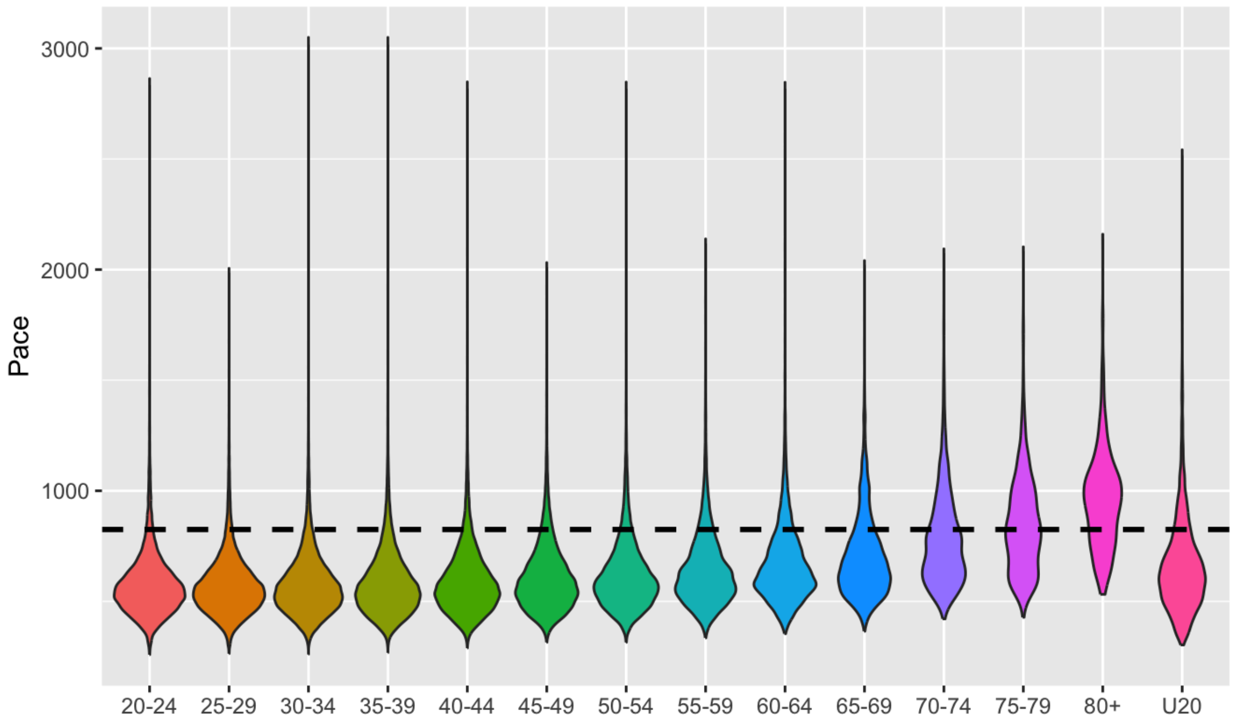

Figure 9: NYRR data by age group with a line marking 13:45 pace

The older age groups have a larger share of participants finishing in slower than 13:45 pace, most notably for the road racers 80 and older, but all age groups have people who finish at paces much slower than 13:45 as evidenced by the long tails on the violin plots.

When we look at the same dataset by gender instead of age, more women are behind that line than men. The nonbinary field is much smaller than the men's and women's fields, but has a wider tail behind the line. The nonbinary field is only 1,661 finishes, so not large enough to draw any specific conclusions, but it suggests that continuing to collect data would be worth doing to get more insight into how nonbinary road racers are affected by these industry standards.

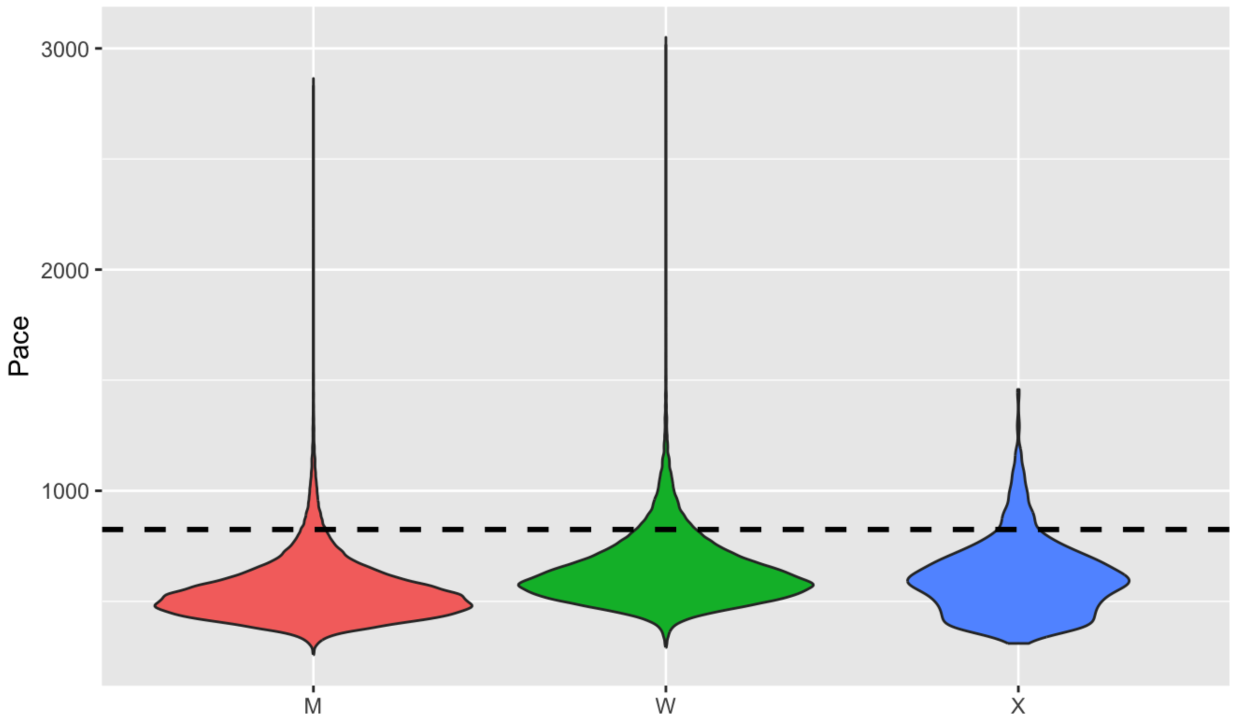

Figure 10: NYRR data by gender with a line marking 13:45 pace

All of this points to one of the key points when looking at the road racing community as it exists now: The back of the pack has fewer runners spread out over more time, leading to far less density on course and a built-in argument against accommodation.

One thing we haven't looked at yet is how Athletes with Disabilities (AWD) fit into the picture. While race registrations don't show AWD status, the Achilles 4M race has more AWD than most races, so is a good point of comparison.

For all 4-mile races, the 25th percentile is 8:13 and the 75th percentile is 10:56 The median pace is 9:24 and the mean is 9:51. When we look just at the 27,114 entries in the Achilles race over the four years, all of those paces slow: The 25th percentile is 8:32 and the

75th percentile is 11:34, almost at the cutoff for the final NYRR corral. That would put a quarter of the participants in this race in a single corral of 12. The median pace is 9:49 and the mean is 10:27. The threshold between Corrals I and J is 9:51, so half the field for this race is lumped into four of 12 corrals.

When we consider the 13:45 pace threshold, a larger share of Achilles race finishes are slower than that compared to all races: 10.7 percent vs. 7.3 percent. As with older road race participants, the current industry standards are more likely to exclude participants with disabilities.

CONCLUSION

After examining this dataset across a number of measures, findings do highlight several places the running industry's standards don't align with actual road racing participation:

- The common beginner running program doesn't meet the needs of about half the road racing community.
- Start line organization pushes too many people of too-varied paces into one group at the back of races.
- Course time limits disproportionally affect older runners, female runners and runners with disabilities.
- The running industry is ill-prepared to allow runners from the first running boom to continue to participate as they age.

Changes outlined in this whitepaper would close the gap between the actual community of road race participants and the people with the power to shape events and training resources.

EVALUATION

My goal with this project was threefold: to build a dataset others could use to analyze a collection of race results that was more inclusive than the running community write large; to assess gaps between industry standards and the reality of road racing participants and participation; and to present my findings in a format that was accessible to runners and others interested in the community. I believe all three of those goals were met, although I also believe there is more work to be done.

The dataset is available for download to anybody with access to GitHub or CUNY Academic Works, which allows them to do their own analysis. I used my journalism background to create a webpage geared to a typical reader rather than an academic, with the analysis distilled into findings that are clear and illustrated with visuals.

The biggest shortcoming with the dataset is that we have limited information on each participant, so the analysis can’t account for running experience, training volume, race/ethnicity, AWD status or socioeconomic status. At the same time, that could be considered a strength because it reduces the arguments detractors would make about *which* finishes should be considered. Crucially, the dataset does not wrestle with whether somebody is running, walking or doing a combination of the two. It treats participation in a road race as sufficient to qualify as a member of the road racing community.

FURTHER RESEARCH

Compiling and cleaning the data took far more time than I had anticipated, so I was unable to incorporate the same four years of data from earlier datasets to look at longitudinal changes in the road racing community, as well as to follow cohorts from the years of the first running boom through to the present, when the 20-year-old runners from 1972-75 are now the 70-year-old runners in 2022-25. That would have allowed a better look at the changes as more women were allowed to participate and as the community went through subsequent running booms and expanded participation.

The past few years have also seen a marked increase in the number of road races affected by weather extremes, and since heat is known to slow running paces, another aspect that should be assessed is how course time limits might need to change to accommodate weather extremes without forcing road racers to push themselves to unsafe paces or to go without fluid stations and medical course support. Twice in recent years, NYRR has had deaths occur at races in unseasonably warm weather, so this is both a safety issue and an accessibility issue.

I also had issues compiling the half-marathon race results because the fields for those races were so large and had to exclude that distance from this analysis. Because the half-marathons run on city streets instead of park loops did not extend the course time limits until a couple of years after 2022, that data would have allowed more examination of a real-time experiment in how extending limits changes the shape of the field of road racers.

As mentioned earlier, this data doesn't reflect the relative experience levels of participants, which limited some of the analysis. Working with NYRR or other race organizations to collect experience levels as part of race registration would allow for a more nuanced look at how these standards impact new runners in older age groups compared to longtime runners who have experience but have slowed with age.

Finally, the policy changes NYRR has might have been offset by increased demand for weekly races by people trying to get guaranteed entry in the marathon, thus shutting out people who don't have the time or internet access to queue online for multiple hours in the middle of the day when race registration opens. Since demand for those race registration slots is unlikely to decline given growing interest in all the World Marathon Majors, I would like to expand the dataset to include more races by other organizations that also allow for the longer finishing times necessary to avoid circumscribing the results and excluding slower runners.

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